

EXPERIMENT N-1

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TITLE : EXPERIMENT N-1 - RADIOACTIVITY DECAY ANALOGUE

RESULTS;

Number of Throws T	Number of dice removed ΔN	Number of dice remaining N	$\ln N$
1	50	250	5.521
2	40	210	5.347
3	39	171	5.142
4	20	151	5.017
5	12	139	4.934
6	17	122	4.804
7	17	103	4.635
8	17	85	4.443
9	12	73	4.290
10	12	61	4.111
11	8	53	3.970
12	7	46	3.827
13	5	41	3.714
14	12	29	3.367
15	5	24	3.178
16	4	20	2.996
17	6	14	2.639



EXPERIMENT N-1

OBJECTIVES

To learn how to simulate Radioactive decay using statistical Process.

THEORY

The Rutherford and Soddy law states that at any instant of time, t the rate of decay of radioactive atoms (dN/dt) is proportional to the number of atoms N present at that instant. As an equation this can be expressed as

$$-\frac{dN}{dt} \propto N$$

$$\text{OR } \frac{dN}{dt} = -\lambda N \quad \text{--- (1)}$$

Where λ is the decay constant and the negative sign means that the number of atoms remaining is decreasing. Note that for a particular element, the decay constant is constant, but varies for different elements. The solution of Equation (1) has the form:

$$N = N_0 e^{-\lambda t} \quad \text{--- (2)}$$

The time taken for half the number of dice N_0 , you are provided with in large box to decay is called half life (T) given by:

$$T = \frac{0.693}{\lambda}$$

LIST OF APPARATUS

Dice / wooden cubes.

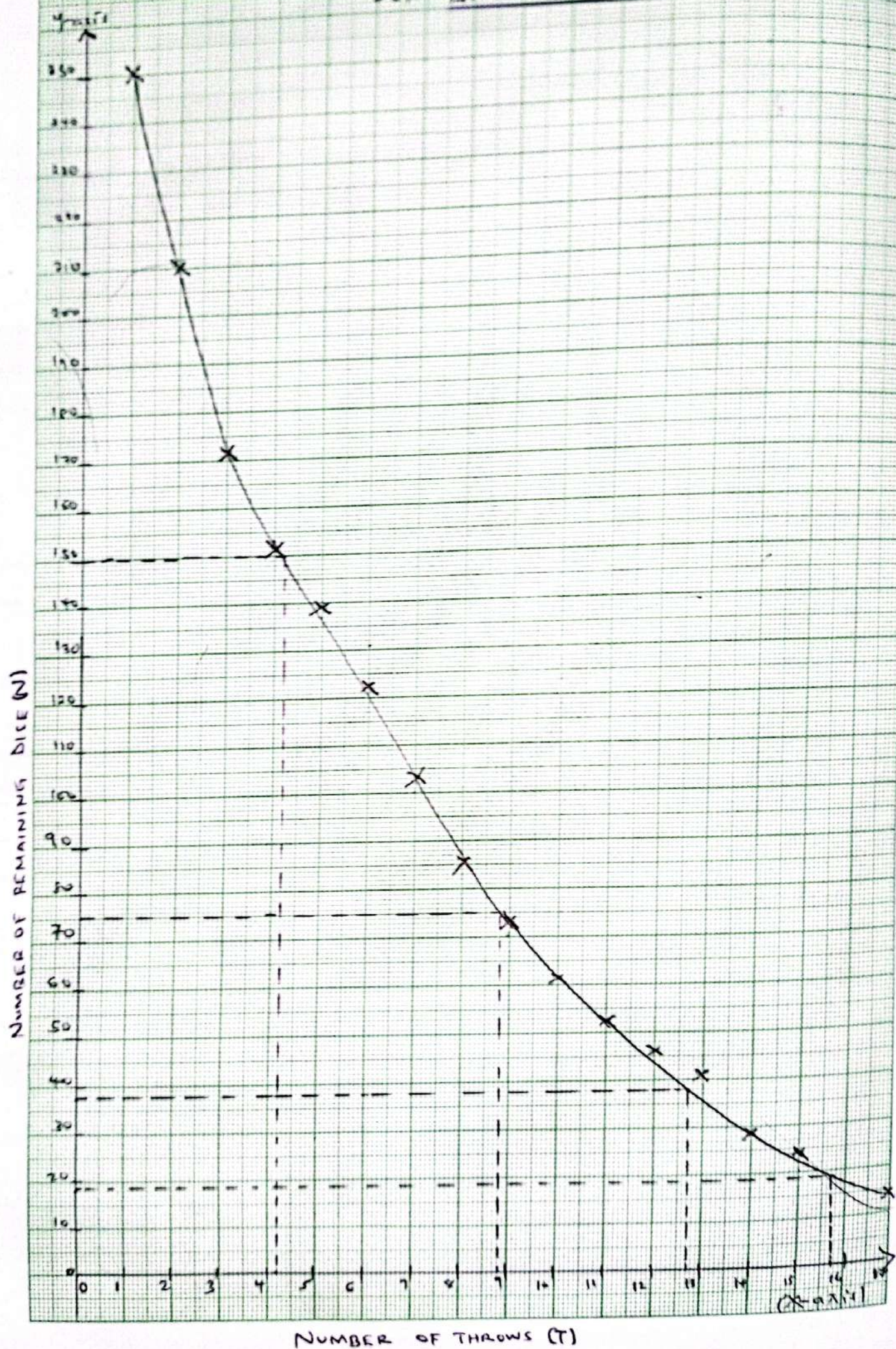
Small box.

Large box.

PROCEDURE

1. Count the no. of dice (wooden cubes) N_0 , you are provided with in the large box
2. Throw all the dice on the table or floor.
3. Remove the dice whose marked faces are uppermost and record their number ($dN = N_0 - N$) against the number of Throws T and put them into the small box. Make your recordings in Table 1
4. Return the remaining dice back into the large box and note the number N . Repeat steps 2 and 3. Repeat until the remaining dice is less than $1/16$ of the original number N_0 .

EXPERIMENT N2
A GRAPH OF (N) AGAINST NUMBER OF THROWS T
FOR ANALYSIS (a)



RESULTS

Analysis:

- a) From the graph determine the number of throws required to have a number of dice decay to

$$N_0/2 \quad N_0/2 = \frac{300}{2} = 150; \quad T = 4$$

$$N_0/4 = \frac{300}{4} = 75; \quad T = 8.8 \approx 9$$

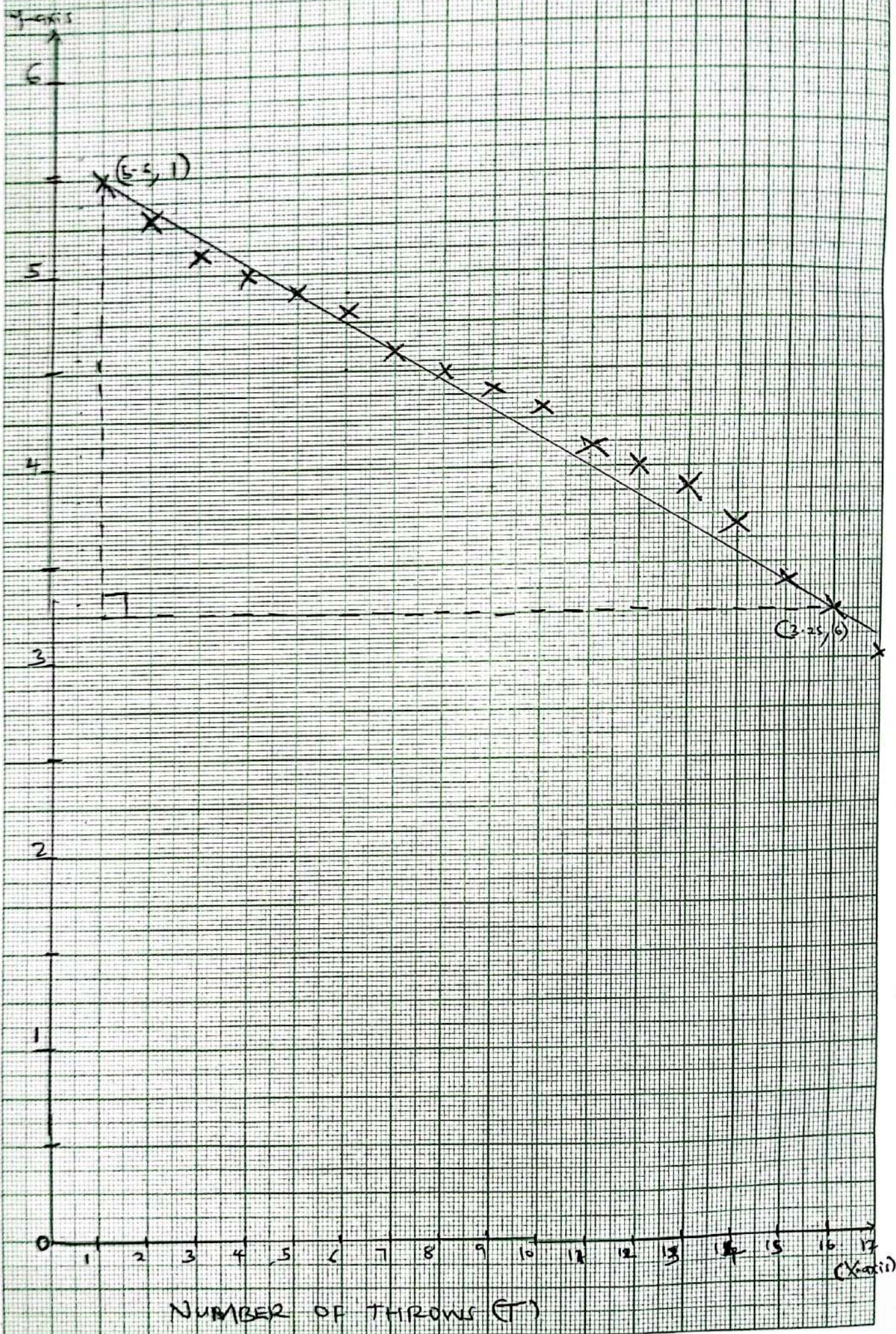
$$N_0/8 = \frac{300}{8} = 37.5; \quad T = 12.7 \approx 13$$

$$N_0/16 = \frac{300}{16} = 18.75; \quad T = 15.8 \approx 16$$

$$\text{Mean } T = \frac{4 + 9 + 13 + 16}{4} = 10.5 \approx 11 \text{ throws.}$$

; Values obtained are approximation of the computed values hence a very minimal difference that is negligible

A GRAPH OF $\ln N$ AGAINST NUMBER OF THROWS T



b) Determine slope of graph.

$$(5.5, 1) ; (3.25, 16)$$

$$\frac{d \ln N}{dt} = \frac{1 - 16}{(3.25 - 5.5) \lambda}$$

$$\text{slope} = -6.667$$

Questions

1. Show that equation (2) satisfies equation 1

$$dN = -\lambda N dt$$

$$\frac{dN}{N} = -\lambda dt$$

$$\int_{N_0}^N \frac{dN}{N} = -\lambda \int_0^t dt$$

$$\ln N - \ln N_0 = -\lambda t$$

$$\ln \left(\frac{N}{N_0} \right) = -\lambda t$$

$$\frac{N}{N_0} = e^{-\lambda t}$$

$$\therefore N = N_0 e^{-\lambda t}$$

2. Derive equation (3) starting with equation (2)

$$N = N_0 e^{-\lambda t}$$

$$\ln \left(\frac{N}{N_0} \right) = -\lambda t$$

$$\ln \left(\frac{1}{2} \right) = -\lambda t$$

$$\frac{-0.693}{-\lambda} = \frac{\lambda t}{-\lambda}$$

$$T = \frac{0.693}{\lambda}$$

3. What is the shape of graph of $\ln$ natural logarithm of equation (2)? Does the slope have any physical meaning.

- The graph is a straight line with a negative gradient
- The slope is equivalent to the logarithm of number of decaying atoms.

4. Suppose you're provided with 500 dice. Calculate the number of throws that is equivalent to one half-life and decay constant.

$$250^{\frac{1}{2}} = 15.81$$

$$T = 16 \text{ throws}$$

$$T = \frac{0.693}{\lambda}$$

$$\frac{16 \lambda}{16} = \frac{0.693}{16}$$

$$\lambda = 0.0433125$$

Discussion of results in comparison to theory.

Results from practical vary from theoretical values; Because radioactive decay is completely spontaneous and random event such that the exact number of atoms decaying cannot be constant in real life hence difference in values.

conclusion

The process of radioactive decay is a spontaneous and random process. The decay from a practical value may slightly vary from the theoretical value.